

CHAPTER 2: FRACTIONS

Exercise 2D Comprehensive Solutions Guide

Rigorous Mathematical Explanations & Pedagogy

1 Multiple Choice Questions

Choose the correct option in each of the following:

Question 1

Which of the following fractions can be expressed as a mixed fraction?

- (a) $\frac{3}{5}$ (b) $\frac{1}{200}$ (c) $\frac{11}{10}$ (d) $\frac{32}{40}$

Mathematical Solution:

- **Definition:** A mixed fraction is represented as $w\frac{a}{b}$, where w is a natural number and $\frac{a}{b}$ is a proper fraction.
- **Rule:** A fraction can be expressed as a mixed fraction if and only if it is an **improper fraction** (i.e., its numerator is greater than or equal to its denominator: $numerator \geq denominator$).
- Let us analyze each option:
 - (a) In $\frac{3}{5}$, $3 < 5$, so it is a proper fraction.
 - (b) In $\frac{1}{200}$, $1 < 200$, so it is a proper fraction.
 - (c) In $\frac{11}{10}$, $11 > 10$, so it is an improper fraction. Converting to mixed form:

$$11 \div 10 = 1 \text{ with a remainder of } 1 \implies \frac{11}{10} = 1\frac{1}{10}$$

- (d) In $\frac{32}{40}$, $32 < 40$, so it is a proper fraction.

Correct Option: (c) $\frac{11}{10}$

Question 2

Which of the following is an improper fraction?

- (a) $\frac{7}{13}$ (b) $\frac{3}{2}$ (c) $\frac{4}{200}$ (d) $\frac{11}{121}$

Mathematical Solution:

- **Definition:** An improper fraction is a fraction whose numerator is greater than or equal to its denominator.

- Let us inspect the options:

- (a) In $\frac{7}{13}$, $7 < 13$, so it is proper.
- (b) In $\frac{3}{2}$, $3 > 2$, so it is improper.
- (c) In $\frac{4}{200}$, $4 < 200$, so it is proper.
- (d) In $\frac{11}{121}$, $11 < 121$, so it is proper.

Correct Option: (b) $\frac{3}{2}$

Question 3

Three-fourths of 3 is:

- (a) a unit fraction (b) a proper fraction (c) an improper fraction (d) not a fraction

Mathematical Solution:

- Mathematically, the operator ”of” translates directly into multiplication:

$$\text{Three-fourths of } 3 = \frac{3}{4} \times 3 = \frac{3 \times 3}{4} = \frac{9}{4}$$

- Let us analyze the resulting fraction $\frac{9}{4}$:
 - In $\frac{9}{4}$, the numerator is 9 and the denominator is 4.
 - Since $9 > 4$, the numerator is greater than the denominator.
 - Therefore, by definition, $\frac{9}{4}$ is an **improper fraction** (which can also be written in mixed form as $2\frac{1}{4}$).

Correct Option: (c) an improper fraction

Question 4

If the cost of a pen is Rs. $16\frac{2}{3}$, then the cost of 60 pens will be:

- (a) Rs. 960 (b) Rs. 480 (c) Rs. 500 (d) Rs. 1000

Mathematical Solution:

- Given, Cost of 1 pen = Rs. $16\frac{2}{3}$.
- First, convert the mixed number to an improper fraction:

$$16\frac{2}{3} = \frac{(16 \times 3) + 2}{3} = \frac{48 + 2}{3} = \frac{50}{3}$$

- Now, find the cost of 60 pens using multiplication:

$$\text{Cost of 60 pens} = 60 \times \text{Cost of 1 pen} = 60 \times \frac{50}{3}$$

- Perform cross-cancellation by dividing 60 by 3:

$$\text{Cost of 60 pens} = \frac{60}{1} \times \frac{50}{3} = 20 \times 50 = \text{Rs. } 1000$$

Correct Option: (d) Rs. 1000

Question 5

If a sheet of paper having an area of $42\frac{1}{4} \text{ cm}^2$ is cut into 13 strips of equal area, then the area of each strip will be:

- (a) $4\frac{1}{4} \text{ cm}^2$ (b) $3\frac{1}{4} \text{ cm}^2$ (c) $4\frac{1}{52} \text{ cm}^2$ (d) $7\frac{1}{13} \text{ cm}^2$

Mathematical Solution:

- Given, Total Area = $42\frac{1}{4} \text{ cm}^2$.

- Convert to an improper fraction:

$$42\frac{1}{4} = \frac{(42 \times 4) + 1}{4} = \frac{168 + 1}{4} = \frac{169}{4} \text{ cm}^2$$

- Since the sheet is divided into 13 equal parts, we perform a division operation:

$$\text{Area of each strip} = \frac{169}{4} \div 13$$

- Apply the reciprocal rule of division:

$$\text{Area of each strip} = \frac{169}{4} \times \frac{1}{13}$$

- Perform cross-cancellation (since $13 \times 13 = 169$):

$$\text{Area of each strip} = \frac{\cancel{169}^{13}}{4} \times \frac{1}{\cancel{13}_1} = \frac{13}{4} = 3\frac{1}{4} \text{ cm}^2$$

Correct Option: (b) $3\frac{1}{4} \text{ cm}^2$

Question 6

Anushka studied these pictures of fractions (representing $\frac{2}{3}, \frac{2}{4}, \frac{2}{5}, \frac{2}{6}$). What pattern might she correctly notice in the fractions?

- (a) Increasing the denominator increases the value of the fraction.
- (b) If the denominator stays the same and the numerator increases, the fraction remains the same.
- (c) Increasing the denominator by 2 cuts the size of the fraction in half.
- (d) If the numerator stays the same and the denominator increases, the fraction decreases.

Mathematical Solution:

- Consider the given series of fractions: $\frac{2}{3}, \frac{2}{4}, \frac{2}{5}, \frac{2}{6}$.
- Notice that the **numerator remains constant** at 2, representing a constant 2 units being selected.
- The **denominator is increasing** (from 3 to 6), which means the whole is divided into more and more equal shares, making each individual share progressively smaller.
- Comparing the decimal values:

$$\frac{2}{3} \approx 0.67 > \frac{2}{4} = 0.50 > \frac{2}{5} = 0.40 > \frac{2}{6} \approx 0.33$$

- Clearly, as the denominator increases while the numerator remains constant, the overall value of the fraction decreases.

Correct Option: (d) If the numerator stays the same and the denominator increases, the fraction decreases.

Question 7

How many pieces of equal size can be cut from a rope of 30 m long, each measuring $3\frac{3}{4}$ m?

- (a) 10 (b) 6 (c) 4 (d) 8

Mathematical Solution:

- Given, Total length of rope = 30 m.
- Length of each piece = $3\frac{3}{4}$ m = $\frac{(3 \times 4) + 3}{4} = \frac{15}{4}$ m.
- Let the number of pieces be N . We divide total length by piece length:

$$N = 30 \div \frac{15}{4} = 30 \times \frac{4}{15}$$

- Perform cross-cancellation:

$$N = \frac{\cancel{30}^2}{1} \times \frac{4}{\cancel{15}_3} = 2 \times 4 = 8 \text{ pieces}$$

Correct Option: (d) 8

Question 8

The value of $3 - \frac{1}{1 + \frac{2}{2 - \frac{3}{5}}}$ is:

- (a) $\frac{37}{17}$ (b) $\frac{38}{17}$ (c) $\frac{39}{17}$ (d) $\frac{44}{17}$

Mathematical Solution:

- Solve step-by-step from the lowermost fraction upward:

- **Step 1:** Simplify the denominator $2 - \frac{3}{5}$:

$$2 - \frac{3}{5} = \frac{10}{5} - \frac{3}{5} = \frac{7}{5}$$

- **Step 2:** Substitute back to evaluate the term $\frac{2}{2 - 3/5}$:

$$\frac{2}{7/5} = 2 \times \frac{5}{7} = \frac{10}{7}$$

- **Step 3:** Simplify the next denominator $1 + \frac{10}{7}$:

$$1 + \frac{10}{7} = \frac{7}{7} + \frac{10}{7} = \frac{17}{7}$$

- **Step 4:** Evaluate the term $\frac{1}{17/7}$:

$$\frac{1}{17/7} = \frac{7}{17}$$

- **Step 5:** Perform the final subtraction:

$$3 - \frac{7}{17} = \frac{3 \times 17}{17} - \frac{7}{17} = \frac{51 - 7}{17} = \frac{44}{17}$$

Correct Option: (d) $\frac{44}{17}$

Question 9

Which of the following represents the shaded parts shown in the figures? (Where the left triangle is fully shaded and the right triangle has 3 out of 4 parts shaded)

- (a) $\frac{2}{4} + \frac{3}{4}$ (b) $\frac{4}{4} + \frac{3}{4}$ (c) $\frac{2}{4} + \frac{1}{4}$ (d) $\frac{1}{4} + \frac{3}{4}$

Mathematical Solution:

- Each large triangle is divided into 4 smaller congruent triangles.
- Left Figure:** All 4 parts are shaded. This represents the fraction:

$$\frac{\text{Shaded parts}}{\text{Total parts}} = \frac{4}{4} = 1 \text{ (whole)}$$

- Right Figure:** 3 out of 4 parts are shaded. This represents the fraction:

$$\frac{\text{Shaded parts}}{\text{Total parts}} = \frac{3}{4}$$

- Combining both figures, the total shaded parts are represented by:

$$\text{Total} = \frac{4}{4} + \frac{3}{4} = 1\frac{3}{4}$$

Correct Option: (b) $\frac{4}{4} + \frac{3}{4}$

Question 10

What is the least fraction that must be added to $(1\frac{1}{3} \div 1\frac{1}{2} \div 1\frac{1}{9})$ to make the result an integer? (Or alternatively, under a plus operator transcription of $(1\frac{1}{3} + 1\frac{1}{2}) \div 1\frac{1}{9}$)

- (a) $\frac{1}{2}$ (b) $\frac{1}{5}$ (c) $\frac{3}{4}$ (d) $\frac{2}{3}$

Mathematical Solution:

- Case A (Strict Division Interpretation):** Evaluate the term $1\frac{1}{3} \div 1\frac{1}{2} \div 1\frac{1}{9}$

- Convert mixed numbers to improper fractions:

$$1\frac{1}{3} = \frac{4}{3}, \quad 1\frac{1}{2} = \frac{3}{2}, \quad 1\frac{1}{9} = \frac{10}{9}$$

- Solve left-to-right:

$$\begin{aligned} \frac{4}{3} \div \frac{3}{2} \div \frac{10}{9} &= \left(\frac{4}{3} \times \frac{2}{3} \right) \div \frac{10}{9} = \frac{8}{9} \div \frac{10}{9} \\ &= \frac{8}{9} \times \frac{9}{10} = \frac{8}{10} = \frac{4}{5} \end{aligned}$$

- The next integer after $\frac{4}{5}$ is 1. The fraction to be added to obtain an integer is:

$$1 - \frac{4}{5} = \frac{1}{5}$$

- **Case B (Addition-First Interpretation):** Evaluate $(1\frac{1}{3} + 1\frac{1}{2}) \div 1\frac{1}{9}$

– Solve brackets first:

$$1\frac{1}{3} + 1\frac{1}{2} = \frac{4}{3} + \frac{3}{2} = \frac{8+9}{6} = \frac{17}{6}$$

– Divide by $1\frac{1}{9} = \frac{10}{9}$:

$$\frac{17}{6} \div \frac{10}{9} = \frac{17}{6} \times \frac{9}{10} = \frac{51}{20} = 2\frac{11}{20}$$

– The next integer after $2\frac{11}{20}$ is 3. The fraction to be added is:

$$3 - \frac{51}{20} = \frac{60-51}{20} = \frac{9}{20}$$

Correct Option: (b) $\frac{1}{5}$ (following Case A)

Question 11

I read $\frac{4}{9}$ of a book on one day and $\frac{3}{5}$ of the remaining next day. If 100 pages of the book were still left unread, how many pages did the book contain?

- (a) 450 (b) 360 (c) 345 (d) 280

Mathematical Solution:

- Let the total number of pages in the book be x .
- **Day 1:** Pages read = $\frac{4}{9}x$.
- Remaining pages after Day 1:

$$\text{Remaining} = x - \frac{4}{9}x = \frac{5}{9}x$$

- **Day 2:** Pages read = $\frac{3}{5}$ of remaining pages:

$$\text{Day 2 Read} = \frac{3}{5} \times \left(\frac{5}{9}x\right) = \frac{3}{5} \times \frac{5}{9}x = \frac{3}{9}x = \frac{1}{3}x$$

- **Total pages read in 2 days:**

$$\text{Total Read} = \text{Day 1 Read} + \text{Day 2 Read} = \frac{4}{9}x + \frac{3}{9}x = \frac{7}{9}x$$

- **Pages left unread:**

$$\text{Unread} = x - \frac{7}{9}x = \frac{2}{9}x$$

- Given that the unread pages equal 100:

$$\frac{2}{9}x = 100 \implies 2x = 900 \implies x = 450$$

Correct Option: (a) 450

Question 12

A drum of petrol is $\frac{3}{4}$ full. When 30 litres of oil are drawn from it, it is $\frac{7}{12}$ full. The capacity of the drum is:

- (a) 200 l (b) 180 l (c) 170 l (d) 150 l

Mathematical Solution:

- Let the total capacity of the drum be C litres.
- Initial volume of petrol = $\frac{3}{4}C$ litres.
- Volume of petrol after drawing 30 litres = $\frac{7}{12}C$ litres.
- According to the problem statement:

$$\frac{3}{4}C - 30 = \frac{7}{12}C$$

- Rearrange the equation to group variables on the left side:

$$\frac{3}{4}C - \frac{7}{12}C = 30$$

- Find the Least Common Denominator (LCD) of 4 and 12, which is 12:

$$\frac{3 \times 3}{4 \times 3}C - \frac{7}{12}C = 30 \implies \frac{9}{12}C - \frac{7}{12}C = 30$$

$$\frac{2}{12}C = 30 \implies \frac{1}{6}C = 30$$

$$C = 30 \times 6 = 180 \text{ litres}$$

Correct Option: (b) 180 l

2 Mental Maths

2.1 Fill in the blanks

1. If $\frac{a}{b}$ is a fraction, then a and b are **natural numbers**.
2. In an improper fraction, the numerator is **greater than or equal to** the denominator.
3. The product of two improper fractions is a/an **improper** fraction.
4. $\frac{a}{b} \div \frac{c}{d}$ equal to the product of $\frac{a}{b}$ and **the reciprocal of $\frac{c}{d}$ (which is $\frac{d}{c}$)**.
5. The product of two proper fractions is **less** than each of the given fractions.
6. The reciprocal of a proper fraction is a/an **improper** fraction.
7. The fraction whose reciprocal is equal to the fraction itself is **1**.

2.2 Write True (T) or False (F)

1. In an improper fraction, the numerator is always greater than the denominator.
False (*Explanation:* The numerator can be equal to the denominator, e.g., $\frac{5}{5} = 1$, which is improper).
2. The product of two proper fractions can be an improper fraction.
False (*Explanation:* Proper fractions are < 1 ; their product is always < 1 and thus always proper).
3. Any improper fraction is always greater than any proper fraction.
True (*Explanation:* Improper fractions are ≥ 1 , whereas proper fractions are < 1).
4. Every unit fraction is equal to 1.
False (*Explanation:* A unit fraction is of the form $\frac{1}{n}$. Only $\frac{1}{1} = 1$. Other unit fractions like $\frac{1}{3}$ are < 1).
5. The reciprocal of a proper fraction is an improper fraction.
True (*Explanation:* If $\frac{a}{b} < 1 \implies a < b$, then its reciprocal is $\frac{b}{a}$, where $b > a$, which is improper).
6. The product of two proper fractions is always greater than each of the two proper fractions.
False (*Explanation:* The product of two numbers less than 1 is smaller than either of them, e.g., $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4} < \frac{1}{2}$).
7. There exists a fraction whose multiplicative inverse is equal to the fraction itself.
True (*Explanation:* The fraction $\frac{1}{1} = 1$ has a reciprocal of 1).
8. The product of a proper fraction and an improper fraction is always less than the improper fraction.
True (*Explanation:* Multiplying a positive quantity by a value less than 1 (the proper fraction) always decreases its value).

3 Case Study Based Questions

3.1 Case Study I: Prabhu Dayal's Will

Prabhu Dayal died and left all his money (totaling Rs. 56,00,000) to his grand-children, consisting of 3 boys and 5 girls. In his will, he insisted that each grand-child must get an equal share of the total amount.

Summary of Givens

Total Amount = Rs. 56,00,000 Number of boys = 3 Number of girls = 5
Total number of grand-children = 3 + 5 = 8.

Questions & Solutions

1. What was the share of each child?

Since there are 8 grand-children, we divide the total money by 8:

$$\text{Share of each child} = \frac{\text{Total money}}{8} = \frac{\text{Rs. } 56,00,000}{8} = \text{Rs. } 7,00,000$$

Correct Option: (d) Rs. 7,00,000

2. What fraction of the money did the girls receive?

There are 5 girls, and each receives Rs. 7,00,000.

$$\text{Total money received by girls} = 5 \times \text{Rs. } 7,00,000 = \text{Rs. } 35,00,000$$

$$\text{Fraction received} = \frac{\text{Girls' share}}{\text{Total money}} = \frac{\text{Rs. } 35,00,000}{\text{Rs. } 56,00,000} = \frac{5}{8}$$

Correct Option: (c) $\frac{5}{8}$

3. How much did the boys receive in total?

There are 3 boys, each receiving Rs. 7,00,000.

$$\text{Total money received by boys} = 3 \times \text{Rs. } 7,00,000 = \text{Rs. } 21,00,000$$

Correct Option: (a) Rs. 21,00,000

4. If one of the girls did not take her share and the money is divided among the remaining grand-children, what is the fraction of the money received by the boys?

If one girl declines, the number of participating grand-children becomes $8 - 1 = 7$.

$$\text{New share of each child} = \frac{\text{Rs. } 56,00,000}{7} = \text{Rs. } 8,00,000$$

$$\text{Total money received by 3 boys} = 3 \times \text{Rs. } 8,00,000 = \text{Rs. } 24,00,000$$

$$\text{Fraction of the total money received by boys} = \frac{\text{Rs. } 24,00,000}{\text{Rs. } 56,00,000} = \frac{3}{7}$$

Correct Option: (d) $\frac{3}{7}$

3.2 Case Study II: Amar the Electrician

Amar is an electrician. He bought $7\frac{1}{2}$ bundles of an electric cable where each bundle had $202\frac{4}{5}$ m of cable.

Summary of Givens

$$\text{Number of bundles} = 7\frac{1}{2} = \frac{15}{2}$$

$$\text{Length of cable per bundle} = 202\frac{4}{5} \text{ m} = \frac{(202 \times 5) + 4}{5} = \frac{1014}{5} \text{ m}$$

Questions & Solutions

1. **Find the total length of the cable purchased by Amar.**

We multiply the number of bundles by the length of one bundle:

$$\text{Total Length} = \frac{15}{2} \times \frac{1014}{5} = \left(\frac{15}{5}\right) \times \left(\frac{1014}{2}\right) = 3 \times 507 = \mathbf{1521 \text{ m}}$$

Correct Option: (b) 1521 m

2. **If the cost of cable is Rs. $6\frac{2}{3}$ per metre, find the amount paid by Amar.**

Price per metre = Rs. $6\frac{2}{3}$ = Rs. $\frac{20}{3}$.

$$\text{Total amount paid} = \text{Total length} \times \text{Price per metre} = 1521 \times \frac{20}{3}$$

Perform cross-cancellation:

$$\text{Total amount paid} = \frac{\cancel{1521}^{507}}{1} \times \frac{20}{\cancel{3}_1} = 507 \times 20 = \text{Rs. } \mathbf{10,140}$$

Correct Option: (c) Rs. 10,140

3. **Amar used $2\frac{1}{2}$ bundles of cable for electric connections in the top floor of the building. What length of cable was used for the top floor?**

Number of bundles used = $2\frac{1}{2}$ = $\frac{5}{2}$ bundles.

$$\text{Length used} = \text{Number of bundles} \times \text{Length per bundle} = \frac{5}{2} \times \frac{1014}{5}$$

Perform cross-cancellation:

$$\text{Length used} = \frac{\cancel{5}}{2} \times \frac{1014}{\cancel{5}} = \frac{1014}{2} = \mathbf{507 \text{ m}}$$

Correct Option: (b) 507 m

4. **Amar cut a length of $13\frac{4}{5}$ m from a bundle and divided the remaining cable of this bundle into pieces of 21 m length each. How many pieces of 21 m did he get from this bundle?**

Length of a single bundle = $202\frac{4}{5}$ m.

$$\text{Remaining cable in this bundle} = 202\frac{4}{5} - 13\frac{4}{5} = (202 - 13) + \left(\frac{4}{5} - \frac{4}{5}\right) = \mathbf{189 \text{ m}}$$

Since the remaining length is divided into pieces of 21 m each:

$$\text{Number of pieces} = \frac{189}{21} = \mathbf{9 \text{ pieces}}$$

Correct Option: (a) 9

4 Assertion-Reason Type Questions**Question 1**

Assertion (A): Reciprocal of an improper fraction is a proper fraction.

Reason (R): Reciprocal is also known as multiplicative inverse.

Mathematical Analysis:

- **Assertion (A):** An improper fraction $\frac{x}{y}$ satisfies $x \geq y$. Its reciprocal is $\frac{y}{x}$. Since $x \geq y \implies y \leq x$, the reciprocal has a numerator less than or equal to the denominator, which is a proper fraction (for $x > y$). Thus, the Assertion is **True**.
- **Reason (R):** By definition, the reciprocal of a number n is its multiplicative inverse $\frac{1}{n}$, because $n \times \frac{1}{n} = 1$. Thus, the Reason is **True**.
- **Explanation Check:** While both are true, the definition of reciprocal as "multiplicative inverse" does not explain the mathematical inequality swap that turns a fraction > 1 into a fraction < 1 . Therefore, R is **not** the correct explanation of A.

Correct Option: Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). (Option 2)

Question 2

Assertion (A): The product of two proper fractions is less than each of the fractions.

Reason (R): For any two fractions $\frac{a}{b}$ and $\frac{c}{b}$, we have, $\frac{a}{b} \times \frac{c}{b} = \frac{ac}{b}$.

Mathematical Analysis:

- **Assertion (A):** Let $\frac{p}{q}$ and $\frac{r}{s}$ be proper fractions (meaning $0 < \frac{p}{q} < 1$ and $0 < \frac{r}{s} < 1$). When we multiply any positive number by a factor less than 1, the result is smaller than the original number. Thus, their product $\frac{p \cdot r}{q \cdot s}$ is smaller than both $\frac{p}{q}$ and $\frac{r}{s}$. E.g., $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$, which is less than both. Thus, the Assertion is **True**.
- **Reason (R):** The rule for multiplying fractions is:

$$\frac{a}{b} \times \frac{c}{b} = \frac{a \times c}{b \times b} = \frac{ac}{b^2}$$

The statement claims the product is $\frac{ac}{b}$, which is incorrect because it fails to multiply the denominators. Thus, the Reason is **False**.

Correct Option: Assertion (A) is true but Reason (R) is false. (Option 3)